

Overview & Features

431 is a type of [martensitic stainless steel](#) with good balance between its strength and toughness. It features good corrosion resistance to aqueous solutions of oxidizing acids, most organic acids and organic salts. Out of heat treatment comes the high mechanical properties. Also, it features better corrosion resistance than 410 and [430](#). 431 type is localized as 1Cr17Ni2 in China, and as JIS SUS431 in Japan.

Since 431 is added with 1.25%-2.5% nickel on the basis of 430, it can replace 430 in some applications like stainless steel panels. However, its corrosion resistance and workability shadow in comparison with [304](#) and [316](#).

431 Applications

- 1) Production of nitric acid and acetic acid;
- 2) Light industry;
- 3) Textile and other industries;
- 4) Parts that require strong toughness and corrosion resistance, like shafts, piston rods, pumps and the like.
- 5) Others.

Tags

Grade 431, Type 431, Inox 431, AISI 431, SUS 431, 1.4057, UNS S43100, Acero 431.

Contact

If you want to buy 431 stainless steel products, please contact the RIGHT supplier - CIVMATS to get the latest price.

Chemical Composition

Data Sheet

Grade

ASTM / AISI	DIN	JIS	GB
431	1.4057	SUS431	17Cr16Ni2

Mechanical Properties

Heat Treatment Temperature	σ_b (Mpa)	σ_s (Mpa)	$\delta\%$	HB	HRB	HV
°C	≥	≥	≥	≤	≤	≤

950-1050 1080 600 14 28540 /